Guillermo López

Full Stack Developer

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Summary

CTO and Full Stack Engineer with 6+ years designing and leading distributed, scalable systems. Specialized in backend, clean architectures (Hexagonal, DDD, CQRS, microservices) and the TypeScript/Node ecosystem, with strong AWS and DevOps expertise. Proven track record building and mentoring engineering teams, modernizing production stacks with zero downtime, and shipping complex projects end-to-end. In the last year, focused on Generative AI, agentic systems, MCP servers and LLM integration for both internal enterprise tools and consumer products.

Experience

CTO & Backend Lead, Vlue Logistics LLC – Miami, FL (Remote) Jan 2023 – present

- Promoted from Backend Lead to CTO; took end-to-end ownership of the entire tech ecosystem
- 4 native mobile apps, management back-office, support dashboard, backend platform, AWS infrastructure and public website.
- Led backend migration to clean architecture (DDD, CQRS, Hexagonal, Repository pattern, polyglot persistence) using NestJS, Kafka, MQTT, Firestore, Redis and ClickHouse.
- Designed real-time tracking and driver positioning systems; integrated Here Maps, Stripe, AppsFlyer and Meta Conversions API.
- Modernized stack and AWS infrastructure progressively without disrupting ongoing development or customer experience.
- Built and mentored a high-performance development team across mobile (iOS/Android), backend and DevOps; redefined workflows under Agile/Scrum and contributed to executive-level strategic decisions.

Full Stack & AI Engineer, Freelance / Independent (Remote) Jan 2021 – present

- **Morphicons** | Created an open-source developer tool for universal SVG icon morphing, now used in production applications and integrated across multiple frontend ecosystems. Reached ~2K GitHub stars and 20K+ weekly npm downloads, with an active community contributing ports, integrations and real-world implementations.
- **ZentrumCare** | Multi-agent system for pediatric healthcare lead-to-clinic matching in the United States, targeting parents of medically complex children. Built around 9+ specialized agents orchestrating the full lead lifecycle, a chat UI agent for intake, a voice agent that confirms each child's condition, matching agents that search the internal clinic network and fall back to live web search, and outreach agents that contact external clinics by phone, email and messaging. Target SLA: convert each lead into a confirmed appointment in under 48 hours. Next.js, Vercel AI SDK, Trigger.dev.
- **mongodb-mcp-vercel** | Read-only MongoDB Model Context Protocol server for serverless deployment on Vercel, exposing database access to LLM agents.
- **phimn.ai** | Internal AI platform for a structural engineering firm to automate code-compliance calculations. Dual-source architecture combining RAG over normative documents with foundational LLM reasoning, plus a WebSearcher subagent and Inngest-based orchestration.
- **Numa365** | React Native mobile application delivered as an independent engineer.
- **Other Work** | Delivered e-commerce, internal tools, B2B platforms and CMS-based websites for international clients across Cuba, the United States, Spain, Germany and Venezuela, including Anjamar and other smaller projects.

Backend & DevOps Consultant (Part-time), Mutter Ventures & InformageStudio – Spain (Remote) Jan 2022 – Dec 2023

- Managed AWS infrastructure and maintained backend codebases for a startup incubator/VC firm and a software consultancy.
- Designed backend solutions for new product features under tight delivery cycles.

Backend Lead, Contapp Digital – Dominican Republic (Remote) Jan 2021 – Jan 2023

- Led backend development of a pioneering automated accounting platform on DR, serving businesses and accountants across iOS, Android and web clients.
- Initial stack ExpressJS, TypeScript and MongoDB with comprehensive Jest test coverage.
- Drove migration to a microservices architecture using NestJS, Redis and RabbitMQ, applying Hexagonal Architecture, DDD and CQRS for scalability and maintainability.

Skills

Languages: TypeScript, JavaScript, Python (jr level)

Backend: NestJS, Express, Hono, Node.js

AI & Agents: LLM Applications, AI Agents, Multi-Agent Systems, MCP Servers, RAG, AI SDKs, Prompt Engineering, Agent Orchestration, Tool Calling, Vector Databases, Local LLMs, AI Workflows

Messaging & Cache: Kafka, RabbitMQ, MQTT, Redis

Databases: MongoDB, PostgreSQL, Firestore, ClickHouse

Frontend & Mobile: React, Next.js, TansTack, React Native, Tailwind, shadcn

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

Architecture: DDD, CQRS, Hexagonal, Microservices, Event-driven, Repository Pattern

Methodologies: Agile/Scrum, technical leadership, code review, mentoring

Languages (spoken): Spanish (native), English (professional working proficiency)

Education

Universidad de Camagüey, BS in Civil Engineering – Camagüey, Cuba Sept 2017 – July 2021

- Graduated with honors. Student Research Award (2019).

Universidad Adolfo Ibáñez, Postgrad. in Communication, Social Networks & Relational Negotiation – Chile (Online) Jan 2020 – Dec 2021

Open Source

- morphicons - Morph any SVG icon into any other with maths, no deps
- translate-kit - npm package to streamline i18n setup and maintenance in Next.js with AI.
- mongodb-mcp-vercel - read-only MongoDB MCP server for serverless deployment on Vercel.
- nextjs-stripe-store - ready-to-use Next.js + Stripe template with all the basics built in.
- blobatar.dev - contributor.
- trigger.dev - one-time contributor.